Same-u mixed branch: composed field result

One composition theorem compiled and passed separate axiom readback. It follows the retained degree2+2 same-u mixed branch from original inputs through extraction and subsequent literal doubling. Main attempt2, controls3 and readback1 exited0; the theorem uses only propext, Classical.choice and Quot.sound.

Inputs are two weighted degree-two representations valid for f=X^5+a3X^3+a2X^2+a1X+a0, four nonzero weights, literal crossed q/r equalities, earlier inverse and equal ordinate guards both false, IsCoprime f f.derivative, and2!=0. No output validity or nonzero extracted ordinate is assumed.

The theorem concludes sourceK!=0; the source-ordered route marker is mixed; extracted point z,w,y are nonzero and validPoint holds; the literal doubled output has nonzero z,w and satisfies validTwo. The route classifier tests support first, inverse second, equal/double22 third, zero-slope failure fourth, then mixed. It uses markers only for the other branches and does not model their payloads. Under the theorem premises, the selected route corresponds to the native same_u_mixed return.

Composition uses reviewed ZeroSlope to exclude the error and crossed scaling to derive normalized support equality and slope difference. WeightedExtract supplies the valid point. GlobalSmooth evaluates the coprimality certificate to establish the local condition. LocalSmooth derives its nonzero ordinate and invokes the reviewed doubling invariant. SourcePoint and WeightedDoubling.output preserve the retained literal field expression trees at source lines903–951 and477–510. No new payload algebra was introduced.

Fixed control: over F5, f=X^5-X+1 has derivative -1 and explicit certificate(A,B)=(0,-1). Inputs a=(0,-16,0,96,2,3), b=(0,-81,54,0,3,2) normalize to u=X²-1 with ordinates1 and X. The complete theorem is applied from original validity and literal guards, yielding mixed route and valid output. Additional controls retain equal-branch precedence, inverse precedence when equal/inverse both hold, failure of global coprimality for f=X^5, and failure of2!=0 in F2. Characteristic-five constants require no division by5.

Retained failures: initial main proof needed final Boolean simplification; first control used decide on noncomputable normalized divisions; second control needed explicit nonzero denominator normalization. These were proof elaboration/control issues, with all prior sources and receipts retained. Final controls include the normalization certificates used by the original-input application.

This is a field-level ordered-branch validity theorem. It does not establish divisor class-sum semantics, machine-word guard reflection, compiler execution or operation costs. The global condition is an explicit input certificate; the theorem is not an implementation for checking coprimality. Independent source-fidelity review is pending. No scientific trial or shared source/cache edit occurred.

Replay by copying SameUComposition.lean,Controls.lean,Readback.lean andcheck.py into a fresh private directory, then run the three modules in that order with fresh tags. Existing pinned dependency packets are bound in sources.json. Each Lean command uses120seconds plus5seconds kill grace, one thread and4096MiB. PDF/SVG/JSON, failed attempts and native receipts accompany this result.